American Computer Science League

2025-26 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

How many ordered quadruples make the following Boolean expression TRUE?

$$\overline{(A + \overline{AB(B + C)})(\overline{CD} + D)}$$

A. 0
B. 4
C. 8
D. 16
E. None of the above

2. Boolean Algebra

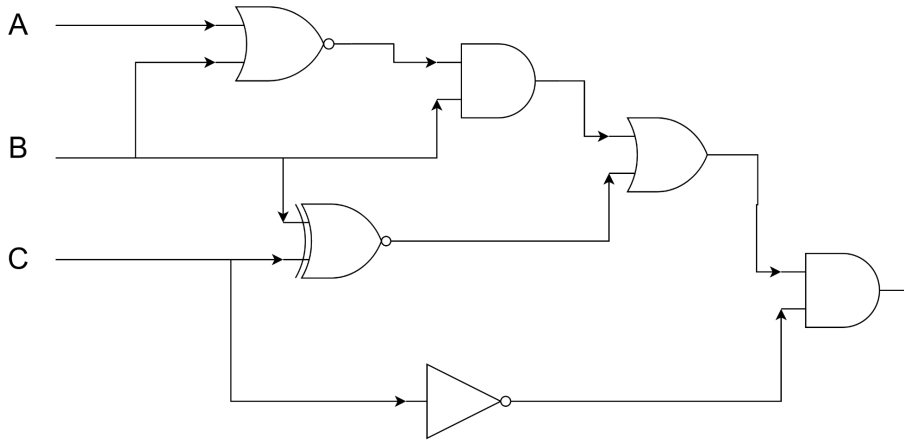
Simplify the following Boolean expression:

$$A(B + C) + \overline{AB} \oplus \overline{C}$$

A. $\overline{A}B + C$
B. $AB + \overline{C}$
C. $AB + C$
D. $\overline{AB} + C$
E. None of the above

3. Digital Electronics

What Boolean expression is a direct translation of the following digital circuit?



A. $((\overline{A + B})B + \overline{B \oplus C}) + C$
B. $((\overline{A + B})B + \overline{B + C})\,\overline{C}$
C. $((\overline{AB} + B)\,\overline{B \oplus C}) + \overline{C}$
D. $((\overline{A + B})B + \overline{B \oplus C})\,\overline{C}$
E. None of the above

4. Bit-String Flicking

Evaluate the following expression by using 8-bit strings and express the answer in octal.

10011101_2 AND (LSHIFT-3 (RCIRC-2 DB_{16}))

A. 100
B. 110
C. 220
D. 230
E. None of the above

5. Bit-String Flicking

Determine what bit strings make the following expression TRUE if X is a 5-bit string:

(RCIRC-2 ((RSHIFT-1 10001) OR X OR (LCIRC-2 10010))) = (LCIRC-3 (NOT 00100))

A. 1*1*1
B. 0***1
C. *1*00
D. 1*0*1
E. None of the above

6. Recursive Functions

Find $f(f(f(f(14))))$, given:

$$f(x) = \begin{cases} f(x-3)+2 & \text{if } x \geq 12 \\ 3 \cdot f(\ \lfloor x \div 2 \rfloor\ ) - 1 & \text{if } 6 \leq x < 12 \\ x-2 & \text{if } x < 6 \end{cases}$$

Note: [y] is the greatest integer less than or equal to y.

A. 2
B. 5
C. 8
D. 10
E. None of the above

7. Recursive Functions

Recursion can be used with numeric and string values. Given the following function using the rules of ACSL pseudo-code, what is the value of f("python")?

$$f(s) = \begin{cases} s & \text{if } len(s) < 2 \\ f(s[len(s)-1:]) + s[0] & \text{otherwise} \end{cases}$$

A. python
B. pnyoth
C. npoyht
D. nohtyp
E. None of the above

8. Prefix/Infix/Postfix Notation

Given that the unary operator % returns the greatest integer less than or equal to its operand divided by 2. evaluate the following prefix expression where all numbers are single digits.

+ / * % 4 + 3 5 % 9 * ^ + ^ 4 2 ^ 3 2 / 1 2 % 5

A. 13
B. 14
C. 16
D. 20
E. None of the above

9. Prefix/Infix/Postfix Notation

Convert the following prefix expression to postfix if all numbers are single digits:

− + ^ 2 3 / * 4 − / 8 2 4 9 / * 6 4 8

A. 23^482/4*−9/+64*8/−
B. 23^482/4−*9/+64*8/−
C. 23^482/4−*9/+64*8−/
D. 23^482/4−*9+/64*8/−
E. None of the above

10. Computer Number Systems This is our 48th year of ACSL. When counting in octal from 1979_{10} to 2026_{10}, inclusive, how many times is the octal digit 0 used?	A. 10 B. 12 C. 14 D. 22 E. None of the above
11. Computer Number Systems The color *dark salmon* is represented by #E9967A. Suppose 6_{16} is added to each component. What is the new hexadecimal color?	A. #EF9C80 B. #EEAC7F C. #F09D80 D. #FFAC7F E. None of the above
12. Data Structures Find the preorder traversal of the binary search tree for: **GOLDSILVERBRONZE**	A. GBDEEOLILNORRSVZ B. BEEDLINOLRRVZROG C. GDBEEOLILONSRRVZ D. BDEEGILLNOORRSVZ E. None of the above

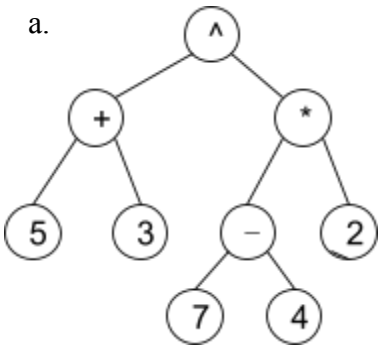
13. Data Structures

A postfix expression can be represented as a binary tree, called an *expression tree*. The following algorithm builds the tree by using a stack of trees:

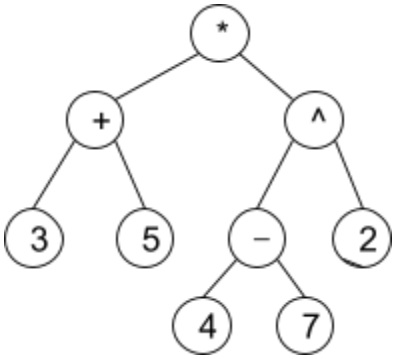
```
for every character ch in the postfix expression
   if ch is a digit
      t = NEWTREE(left=nil, root=ch, right=nil)
      PUSH(t)
   else
      y = POP()
      x = POP()
      t = NEWTREE(left=x, root=ch, right=y)
      PUSH(t)
```

Which of the following will be the final result of the binary tree for the expression 5 3 + 7 4 − 2 ^ *?

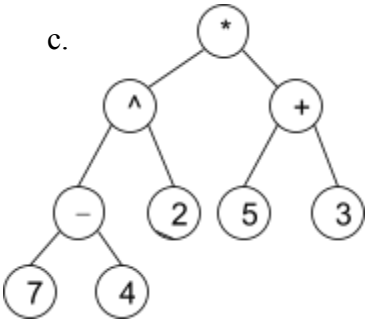
a.



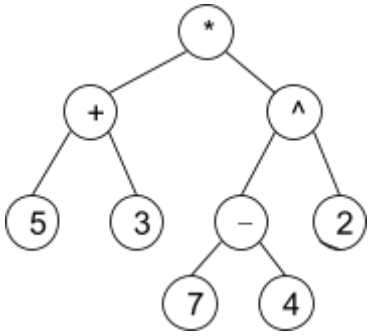
b.



c.



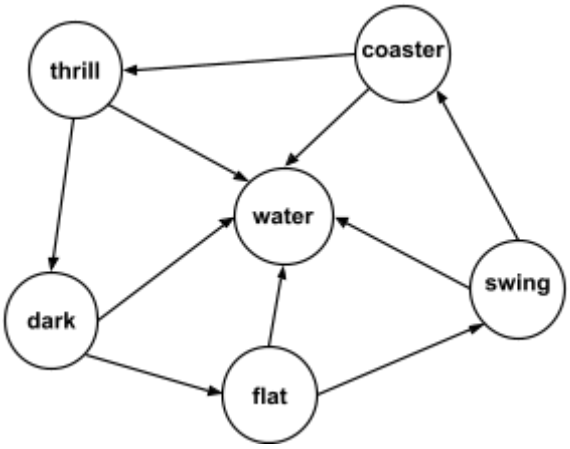
d.



A. a
B. b
C. c
D. d
E. None of the above

14. Graph Theory

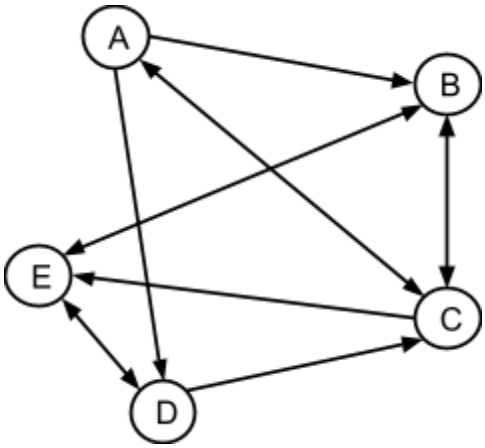
A group of children entered an amusement park. Each child will go on 3 different rides. All of the 3 rides must follow a path in the graph below. How many different paths are possible?



A. 8
B. 9
C. 10
D. 11
E. None of the above

15. Graph Theory

How many cycles in the following graph contain all 5 vertices?



A. 1
B. 2
C. 3
D. 4
E. None of the above

16. What Does This Program Do?

Given that array **arr** has **n** items, the following program uses the Insertion Sort to put values in the array into ascending order. Fill in the box so that it works correctly given that the first item is stored in location 0.

```
for j = 1 to n-1
    key = arr(j)
    k = j - 1
    while k >= 0 && key < arr(k)
        [                    ]
        k = k - 1
    end while
    arr(k+1) = key
next j
```

A. `arr(k)=arr(k-1)`
B. `arr(k)=arr(k+1)`
C. `arr(k+1)=arr(k)`
D. `arr(k-1)=arr(k)`
E. None of the above

17. LISP Programming

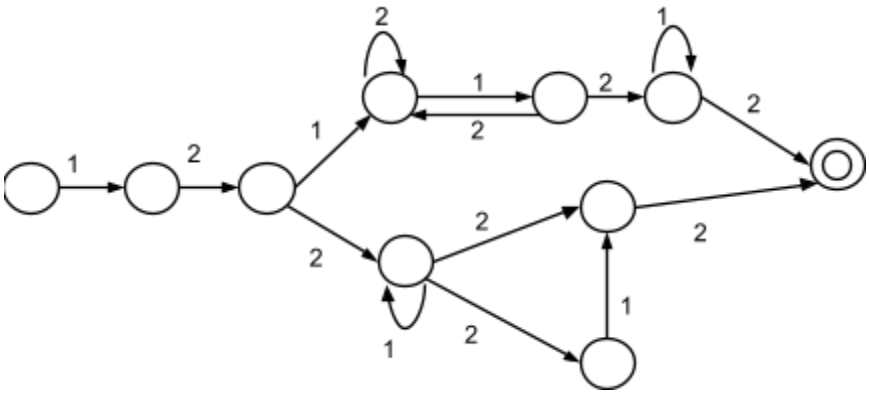
Evaluate the following LISP expressions:

(SETQ X '((a (b b r a) (c a)) ((d a) (b r a))))
(REVERSE (CONS (CDADAR X) (CADADR X)))

A. ((a r b) (b r a))
B. (a r b b r a)
C. (a r b (a r b))
D. (a r b (b r a))
E. None of the above

18. FSAs and Regular Expressions

Which of these strings are accepted by the following FSA?



a. 1211222122
b. 1222112
c. 121122
d. 1221111111212
e. 121221221212
f. 12211121

A. a, b, d, f
B. c, d, e, f
C. a, c, d, e
D. b, c, d, f
E. None of the above

19. FSAs and Regular Expressions

Which of the following strings are accepted by the regular expression:

[^g-r] [^aeiou] [aei] [i-t]* [a-g]* [s-x]*

a. skiing	b. bobsleigh	c. slalom
d. curling	e. hockey	f. snowboard
g. freestyle	h. skating	

A. c, g, h
B. a, c, h
C. a, f, h
D. a, f, g
E. None of the above

20. Assembly Language

What is the output when the following program is executed?

```
NUM       DC          2026
INIT      DC          0
COUNT     DC          1
TOP       LOAD        INIT
          STORE       T
LOOP      LOAD        NUM
          BE          CONT
          DIV         =10
          STORE       QUOT
          MULT        =10
          SUB         NUM
          MULT        =-1
          STORE       D
          MULT        D
          ADD         T
          STORE       T
          LOAD        QUOT
          STORE       NUM
          BU          LOOP
CONT      LOAD        COUNT
          ADD         =1
          STORE       COUNT
          LOAD        T
          STORE       NUM
          SUB         =1
          BE          STOP
          BU          TOP
STOP      PRINT       COUNT
          END
```

A. 5
B. 6
C. 7
D. 8
E. None of the above